

Fadi Atieh

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EDUCATION

Ph.D. in Machine Learning

Starting Fall 2026

Carnegie Mellon University, Pittsburgh, PA

Advisor: Max Simchowitz & Nicholas M. Boffi

Funded by U.S. Department of Energy Computational Science Graduate Fellowship (DOE CSGF)

Masters of Engineering (GPA: 5.0/5.0)

09/2020 - 02/2022

Massachusetts Institute of Technology, Cambridge, MA

Thesis: A Novel Statistical Procedure Toward the Discovery of the Higgs Boson

Mathematics & Computer Science (GPA: 5.0/5.0)

09/2016 - 06/2020

Massachusetts Institute of Technology, Cambridge, MA

ADVANCED COURSEWORK

- Measure-Theoretic Probability • Non-Asymptotic Statistics • Information Theory
- Probabilistic Graphical Models • Mathematical Optimization • Theory of Computation
- Algebraic Combinatorics • Abstract Algebra • Chaos Theory • Real Analysis • Distributed Systems

WORK EXPERIENCE

ML Research under Prof. Aditi Krishnapriyan, UC Berkeley

06/2025 – present

Independent Researcher

- Co-authored work showing Transformers predict molecular energies and forces without graph, symmetry, or physical priors, with errors competitive with SOTA GNNs.
- Integrated the Muon optimizer into FSDP distributed training for 7B-parameter models, implementing custom scatter/gather protocols to handle Muon's 2D gradient matrix materialization across sharded parameter states.
- Showed attention scores decay with inter-atomic distances; defined an “effective attention radius” and showed it adapts to local atomic density, revealing a learned molecular topology.

Citadel LLC

02/2022 - 09/2024

Quantitative Research Engineer at GQS (Global Quantitative Strategies)

- Developed and deployed quantitative models predicting various market variables working closely with quantitative researchers.
- Built a scalable sampling/fitting/inference pipeline for niche models with unique mathematical formulations that are not amenable to preexisting infrastructure.
- Built a scalable model evaluation framework for analyzing research models with plug-and-play metrics of a user's choice.
- Built infrastructure for serving live predictions of various market variables to be used in live trading.
- Consistently rated a top performer within team and cohort via internal evaluations and structured third-party feedback processes.

Citadel LLC

06/2021 - 08/2021

Research Engineering Intern at GQS (Global Quantitative Strategies)

- Optimized the main research file system through changes in file representation and C++ API reducing I/O hits, latency, and storage by an order of a magnitude.
- Analyzed external-source market data for price and volume accuracy reconciling with in-house data.
- Built an automated testing/linting Bazel framework for Python scripts within a large code base.

Citadel LLC

07/2020 - 08/2020

Software Engineering Intern at Commodities

- Built a Flask server used by traders to mark futures curves in commodities and achieved a 100× speed up from the legacy server.
- Exposed C++ instrument valuation functionality into Python using Pybind11 and wrote documentation of the exposed interface.

PUBLICATIONS	Kreiman, T., Bai, Y., Atieh, F. , Weaver, E., Qu, E., Krishnapriyan, A. S. (2025). Transformers Discover Molecular Structure Without Graph Priors. <i>arXiv:2510.02259</i> .	
	Roques-Carmes, C., Shen, Y., Zanoci, C., Prabhu, M., Atieh, F. , Jing, L., Dubček, T., Mao, C., R. Johnson, M., Čeprić, V., D. Joannopoulos, J., Englund, D., Soljačić, M. (2020). Heuristic Recurrent Algorithms for Photonic Ising Machines. <i>Nature Communications</i> .	
RESEARCH EXPERIENCE	<i>Master's Thesis under Prof. Yury Polyanskiy</i>	09/2020 - 02/2022
	Graduate Researcher at Prof. Yury Polyanskiy's Group - Developed a novel hypothesis testing method toward the discovery of the Higgs boson, addressing limitations of Kaggle's HiggsML challenge. - Trained a neural net to approximate the log-likelihood ratio of Higgs decays, integrated into a non-parametric Wilcoxon rank-sum test. - Supported the method with theoretical analysis and validation on LHC data and MC simulations.	
	<i>Photonics & Modern Electro-Magnetics Group under Prof. Marin Soljačić</i>	02/2018 - 12/2018
	Undergraduate Researcher at Research Laboratory of Electronics at MIT - Demonstrated stochastic convergence of a probabilistic Ising solver via simulations and analysis, leading to a peer-reviewed Nature Communications publication. - Implemented fast matrix multiplication for MCMC using VHDL on FPGA, leveraging parallelism.	
PROJECTS	<i>Arabic PL-BERT: Phoneme-level BERT Training Pipeline</i>	02/2025 - 04/2025
	Personal Project - Trained a phoneme-level BERT model from scratch on Arabic Wikipedia for downstream speech tasks. - Preprocessed text using normalization, diacritization (Catt), and phonemization; 1.2M articles for pretraining, 200K for finetuning. - Retained MLM objective without tokenization, improving phonemization accuracy. - Open-sourced the trained model on Hugging Face.	
	<i>Arabic TTS System: Dataset Construction + StyleTTS2 Training</i>	02/2025 - 04/2025
	Personal Project (in collaboration with a peer) - Built a 100+ hr Arabic TTS dataset using denoising, diarization, and Whisper transcription. - Fine-tuned StyleTTS2 on 22 hrs of audiobook data using the Arabic PL-BERT model. - Adapted StyleTTS2 for Arabic (e.g., removed English-specific WavLM loss) and trained via Modal. - Open-sourced the model and demo on Hugging Face.	
	<i>Neural Network Library in Rust</i>	04/2024
	Personal Project - Built a neural network library in Rust with a Python interface for training and inference. - Designed a clean API and demonstrated functionality on synthetic data in Jupyter.	
	<i>Tic-Tac-Toe MCTS (Monte Carlo Tree Search) Game Agent in Rust</i>	03/2024
	Personal Project - Implemented a tic-tac-toe agent using Monte Carlo Tree Search (MCTS) in Rust. - Learned Rust during the project and built a CLI for interactive gameplay.	
MATHEMATICS OLYMPIAD	- Honorable Mention in the Asian Pacific Mathematics Olympiad XXV & XXVI. - Bronze Medal in the Syrian National Math Olympiad Finals.	2013 & 2014 01/2013
TEACHING EXPERIENCE	<i>Theory of Computation with Professor Michael Sipser (18.404)</i>	Fall 2020 & Fall 2021
	Teaching Assistant - Led weekly recitations and collaboratively solved problem sets with students to deepen lecture concepts. - Co-hosted lectures on Zoom, fielding live student questions alongside faculty and other TAs. - Held weekly office hours; received average student ratings of 6.7 and 6.8 out of 7.	